

ANDRE CASAGRANDE ROSENBERG

Artificial Intelligence Graduate Student | Civil Engineer | Applied AI & Computer Vision

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PROFESSIONAL SUMMARY

Master in Artificial Intelligence student at Atlantis University with hands-on academic experience in machine learning, computer vision, Python, PyTorch, torchvision, Jupyter, and MobileNetV2 transfer learning. Civil engineer with professional experience leading engineering, aerial-mapping, photogrammetry, GIS, and geospatial data workflows from field acquisition through spatial analysis and technical delivery.

EDUCATION

Master in Artificial Intelligence - Atlantis University

In progress | Started 2025 | Expected completion: July 2027

Relevant coursework: Machine Learning; Advanced Applied Machine Learning; Natural Language Processing; Artificial Intelligence for Human-Computer Interaction; Ethics in AI; Artificial Intelligence (current).

Bachelor of Civil Engineering - Universidade Vila Velha

Completed August 2020

SELECTED AI PROJECTS

Concrete Crack Image Classification

- Built an academic computer-vision workflow using MobileNetV2 transfer learning to classify concrete surface images as Negative or Positive for visible cracks.
- Used a 40,000-image dataset with an 80/20 train/test split in Python/Jupyter with PyTorch and torchvision.

AI-Assisted Safety Information

- Developed an academic decision-support concept that uses AI to organize and synthesize authoritative safety and reference information while keeping final safety judgment with a human user.

PROFESSIONAL EXPERIENCE

Owner & Responsible Civil Engineer - BR Solos Engenharia Ltda. | June 2023 - Present

- Lead end-to-end engineering and aerial-mapping workflows from field data capture through processing, spatial analysis, cartographic outputs, and technical deliverables.
- Operate aerial imaging and data-acquisition systems and develop engineering projects supported by geospatial information.

Civil Engineer - Autovia Engenharia Ltda. | Aug 2020 - May 2021; Dec 2022 - Aug 2023

- Acquired aerial imagery and field data, processed datasets, and produced maps and technical charts for engineering projects.
- Integrated survey, mapping, and technical-documentation workflows.

Partner & Responsible Civil Engineer - T&R Engenharia Ltda. | Apr 2021 - Jan 2023

- Led engineering responsibility and UAV/aerial data-acquisition activities for mapping, cartography, and project development.
- Converted field imagery and measurements into geospatial products for engineering analysis and planning.

TECHNICAL SKILLS

Python | Jupyter | PyTorch | torchvision | Computer Vision | MobileNetV2 | Machine Learning | UAV / Aerial Imaging | Photogrammetry | GIS / Geoprocessing | Cartography | Spatial Databases | Civil Engineering

PUBLICATION & PRESENTATION

- Publication: *Estudo de Ocorrencias Patologicas em Pontes com a Utilizacao de Veiculos Aereos Nao Tripulados* - April 2020.
- Presentation: *Integracao da Aerofotogrametria e Modelagem 3D na Urbanizacao e Prevencao de Catastrofes Naturais* - Universidade Vila Velha, September 2024.

LANGUAGES

Portuguese | English